



Effects of traditional Chinese herbal medicines on egg production, egg quality and antioxidant enzyme activities in aged laying hens

Chunhong Li^{1*}, Yulong Dong^{1*}, Zhong Hao Li^{1*}, Ruihua Zhang^{1*}, Ruiling Qin^{1*}, Li Wang^{1*} and Wanyu Shi^{2**}

Department of Animal Medicine,
Hebei North University, Zhangjiakou 075131, China.

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ABSTRACT

A study was conducted to evaluate the effect of traditional Chinese herbal medicines (TCHM) on egg production, egg quality and antioxidant enzyme activities in chickens, astragalus membranaceus (AM), angelicae sinensis extract (ASE) and danggui buxue san (DBS) were used for the study. 300 hens of 56 weeks of age were randomized into 10 groups. AM, ASE and DBS were added into the diets of the animals at the dosage of 5mg/Kg, 10 mg/Kg and 15 mg/Kg respectively from group I to group IX. The hens of group X, as of control, were fed with the diet without TCHM. The added TCHM Period was 15 days. The results showed that DBS had the ability on improving egg production, egg quality and antioxidant enzyme activities in chickens. At the same time, DBS was added into the diets of the animals at the dosage of 10mg/Kg was recommended for application.

Key words: Angelicae sinensis extract, Antioxidant enzyme, Astragalus membranaceus, Chickens, Egg production, Egg quality.

INTRODUCTION

Antibiotics can improve health and productive performance of animals, but also result in the development of drug-resistant microorganisms, which can then pass the resistance on to infectious microorganisms in humans (Li *et al.*, 2006). The European Union has banned the use of antibiotics as growth-promoting agents in the poultry industry, and many countries are restricting the use of antibiotic as growth-promoting agents. Therefore, it is essential to find possible alternatives to antibiotics for growth promotion and improvement in poultry production. Thus traditional Chinese veterinary medicines which have no toxicity and less drug residues are becoming more and more popular.

Studies showed that “Blood” and “Qi” deficiency was inevitable in aged animals, because of physiological imbalances. The main manifestations were pale tongue, sparse hair, dullness, thready-weak pulse, decreased egg production and laying rate (Liang *et al.*, 1999). It is necessary to adjust “Blood” and “Qi” for aged animals by traditional Chinese veterinary medicine.

Danggui Buxue Tang (DBT) is an ancient traditional Chinese herbal formula composed of Danggui (Radix Angelica sinensis) and Huangqi (Radix Astragali) with a weight ratio of 1:5 (Gao *et al.*, 2007). DBT was first described in *Neiwaishang Bianhuo Lun* by Li Dongyuan in China in AD 1247. According to Chinese medicinal theory,

the daily intake of DBT could raise the “Qi” and nourish the “blood” of menopausal women. Pharmacological actions indicated that DBT had the effects of increasing anti-oxidation activity, promoting hematopoietic functions, preventing osteoporosis, stimulating cardiovascular circulation, stimulating immune response (Jin and Fereshteh, 2012; Dong, 2006; Gao, 2007). According to free radical theory, decrease in cell physiological function would lead to decrease in egg production, because of raising malondialdehyde (MDA) and decrease in superoxide dismutases (SOD) and glutathione (GSH) in aged animals.

In addition, angelicae sinensis extract (ASE) and Astragalus membranaceus (AM) are commonly used in treating the age-related diseases, which have been demonstrated in increasing bone formation, stimulating bone cell proliferation and reducing bone re-sorption in patients (Yang *et al.*, 2002). APS is a bioactive chemical in Astragalus membranaceus which has been used in medicine for centuries in China and has the effects of anti-oxidation, exhibiting immune-stimulatory, anti-tumor and anti-viral (Dang *et al.*, 2004; Li *et al.*, 2008; Yuan *et al.*, 2008; Sima *et al.*, 2012). AS has biological functions such as antioxidation, aging and immunoregulation (Shao *et al.*, 2006; Cho and Leung, 2007). Angelica polysaccharide had been reported to possess anti-tumor effects (Shang *et al.*, 2003; Tsai *et al.*, 2005) and also exhibit immunostimulating activities both in vitro and in vivo (Cho and Leung, 2007) in cancer cells.

*Corresponding author's e-mail: shiwayuanimal@126.com

¹Department of Animal Medicine, Hebei North University, Zhangjiakou 075131, China.

²College of Veterinary Medicine, Agricultural University of Hebei, Baoding 071001, Hebei, P R China.

**These authors contributed equally to this work.

Previous research showed DBT or Danggui Buxue San (DBS) had effects on raising the “Qi” and nourishing the “blood” in mice and poultry with “blood-Qi” deficiency (Li *et al.*, 2013). However, reports regarding its effects on increasing anti-oxidation activity, improving egg production and egg quality are scanty in aged poultry. In this study, the effect on anti-oxidation activity, egg production and egg quality of AM, ASE and DBS in chickens, the best way and dosage were selected and the mechanisms on increasing anti-oxidation activity were studied.

MATERIALS AND METHODS

Treatment of animals: 56 weeks of age Hyline Brown laying hens were purchased from Hebei laboratory animal center, and were housed in cages. The hens were given free access to feedstuff and water. All the experimental animals were treated in accordance with the guidelines of the Chinese Council for Animal Care.

Preparation of reagents: AM, ASE were purchased from Beijing Biochem Co., Ltd. (China). AM and ASE were smashed respectively to 200 mesh sieve. DBS consisted of ASE and AM in a 1: 5 ratio.

Experiment design: Three hundred 56-week-old normal hens were randomly divided into ten groups with the same number and similar body weight. The hens in group I, II and III were given three gradient dosages (5 mg/Kg, 10 mg/Kg, 15 mg/Kg) of AM respectively. Hens in group VI, V and VI were given three gradient dosages (5 mg/Kg, 10 mg/Kg, 15 mg/Kg) of ASE respectively. In group VII, VIII and IX hens were given DBS by the same gradient dosages respectively. Hens in group X were fed basal diet as control. The trial was continued for 15 days in groups. Hen-day egg production, egg weight, feed consumption and hen mortality were recorded daily. Feed conversion ratio was calculated as grams of feed intake per gram of egg mass.

At 8, 15, 22, 30 days of test, two eggs of each group per replication were collected to measure the egg quality parameters. Haugh units, yolk color, eggshell thickness and

eggshell strength were measured with a digital egg tester after eggs were weighed and cracked open.

Haugh unit= $100 \times \log (H-1.7 \times W \times 0.37+7.6)$

H is egg white height, W is egg weight.

Water content of egg yolk= Water / egg yolk

Blood specimen collection and examination: At 8 and 15 days of the test, blood of 1 hen of each group per replication was collected by wing-vein puncture. Blood samples containing coagulant and centrifuged for 10 min ($3,000 \times g$) at 4°C. Pure serum samples were aspirated by pipette, stored in sterilized 1.5-mL Eppendorf tubes at -80°C, and thawed at 4°C before antioxidant enzyme activities analysis. The GST was measured by ELISA with commercial kits provided by Hangzhou Nuoyang Biotechnology Co. Ltd. (Hangzhou, China), whereas SOD (Winterbourn *et al.*, 1975), GSH (Zakowski and Tappel, 1978), CAT (Beers and Sizer, 1952), and MDA (Gomez *et al.*, 1998) levels were measured by spectrophotometric methods using a spectrophotometer.

Statistical analysis: Data was analysed using SPSS 16.0 statistical software. Statistical significance was determined by one-way analysis of variance (ANOVA) followed by student's *t*-test. A probability of less than 0.01 was considered to be statistically significant.

RESULTS AND DISCUSSION

Effects of chinese traditional herbal medicines on egg production: Table 1 shows that DBS in group VIII (10 mg/Kg) had significant effects on the laying rate, daily-egg weight and feed conversion rate than group X ($P < 0.01$). In addition, the feed cost in group VIII (10 mg/Kg) was much lower than that in group X. DBS (15 mg/Kg) had significant increase on the laying rate, feed conversion rate ($p < 0.05$) and daily-egg weight than group X ($P < 0.01$), but had no more effect on other indexes ($p > 0.05$). AM shown in Table 1, the laying rate and daily-egg weight in group III (15 mg/Kg) had significant effects than group X ($p < 0.05$), but had no more effect on other indexes ($p > 0.05$). There was no more change on egg production in groups IV to VI than the control group

Table 1: Effects of traditional Chinese herbal medicines on egg production index in aged laying hens

Group	N	Dosage (mg/Kg)	Average egg weight (g)	Laying rate (%)	Daily-egg weight (g/d)	Feed intake (g/d)	Feed conversion rate (%)
(I) Low dosage of AM	30	5	65.80±1.17	68.55±5.10	44.30±2.73	123.88±11.49	2.78±0.37
(II) Middle dosage of AM	30	10	65.44±1.55	69.37±4.52	49.13±2.66	124.27±12.12	2.72±0.31
(III) High dosage of AM	30	15	65.83±2.29	71.43±3.95*	46.11±2.31*	124.95±11.15	2.75±0.43
(IV) Low dosage of ASE	30	5	65.35±1.08	68.01±5.50	44.22±2.91	126.21±12.44	2.83±0.28
(V) Middle dosage of ASE	30	10	65.39±1.43	69.07±4.52	45.19±2.77	125.58±11.12	2.80±0.30
(VI) High dosage of ASE	30	15	65.42±2.11	69.36±3.93	45.83±2.53	125.03±12.09	2.79±0.40
(VII) Low dosage of DBS	30	5	65.82±1.10	70.23±5.60	44.33±2.63	123.07±11.38	2.78±0.37
(VIII) Middle dosage of DBS	30	10	65.41±1.48	74.20±4.82**	49.33±2.89**	124.16±10.12	2.53±0.28**
(IX) High dosage of DBS	30	15	65.78±2.25	71.64±4.01*	47.00±2.11**	124.62±12.47	2.67±0.40*
(X) control	30	0	65.35±1.47	68.21±3.14	44.00±3.02	126.95±14.66	2.89±0.40

**Superscript differs significantly ($P < 0.01$) compared with group X,

*superscript differs notably ($P < 0.05$) compared with group X.

($P > 0.05$). In a word, the effects of DBS (10 mg/Kg) on egg production were better than other groups.

Effects of chinese traditional herbal medicines on egg quality: DBS in group VIII (10 mg/Kg) had significant effect on increase of yolk color, the eggshell thickness, eggshell strength and Haugh Unit ($p < 0.05$), decrease in the water content of egg white than group X ($p < 0.05$) (Table 2). DBS in group IX (15 mg/Kg) could significantly increase the Yolk color, eggshell thickness, eggshell strength and decrease the water content of egg yolk than group X ($p < 0.05$). DBS in group VII (5 mg/Kg) could significantly improve the eggshell thickness and eggshell strength than group X ($p < 0.05$). However, there was no more effect on the egg quality in the groups of AM and ASE than the control group ($P > 0.05$), low or high dosage of DBS were not suitable.

Effects of chinese traditional herbal medicines on antioxidant enzyme activities: Supplementation with DBS (10 mg/Kg) significantly improved GST, GSH activity and decreased MDA activity than the control ($p < 0.05$). DBS (5 mg/Kg) also decreased MDA activity over the control ($p < 0.05$). There were no differences for other antioxidant enzymes among all other groups than the control ($p > 0.05$). This indicated that the different dosages of single AM and ASE had no notable effects on the antioxidant enzyme activities, but DBS (10 mg/kg) could improve the antioxidant enzyme activities (Table 3).

The results of this study clearly demonstrated that DBS (10 mg/Kg) could not only increase laying rate, day-egg weight, feed conversion rate, yolk color, eggshell thickness, eggshell strength, Haugh Unit, GSH activity but also decrease feed cost, water content of egg yolk and MDA activity (Table 1, 2 and 3). Only AM or ASE had no more effect. At the same time, low or high dosage of DBS were not suitable.

Studies showed that syndrome of “blood” and “Qi” deficiency was inevitable in aged laying hens, because of physiological imbalances. As the laying week increases, primordial and growing follicles are gradually atrophied,

but atretic follicles increase in ovarium, so the laying rate will decrease (Palmer and Bahr, 1992). Chickens are liable to diseases. Therefore, it is necessary to adjust “Blood” and “Qi” for aged laying hens. Previous studies showed that there was significant effect of DBS (10 mg/Kg) on enriching “blood” and nourishing “Qi”. There are biological functions such as antioxidation, aging and immunoregulation with AM (Shao *et al.*, 2006; Cho and Leung, 2007). The results indicated that the laying rate, day-egg weight and feed conversion were notably increased by DBS (10 and 15 mg/kg) compared with the control group. However, there was no effect on the egg production with AM or ASE. These effects may be related to improved “blood” and “Qi” in laying hens. According to free radical theory, the effect of cell physiological function decrease led to decrease in egg production, because of raising MDA and decreasing SOD and GSH for aged animals.

Some dietary levels of DBS improved egg quality, but AM or ASE had no significant effect (Table 2). Compared with the control, Haugh unit was significantly increased and decrease the Water content of egg Yolk in DBS (10 mg/Kg and 15 mg/Kg) groups. This effect may be related to increased protein synthesis and the transfer of water from yolk. Haugh unit is an important index to measure the quality of egg protein. The value is greater, the quality is better. The thickness and strength of egg shell were significantly increased in all DBS groups compared with the control. That may be associated with the ability of probiotics to decrease pH and improve intestinal barrier function (Resta-Lenert and Barrett, 2003). Lower pH and intestinal epithelium functions both are required for the dissolution of calcium and phosphorus to promote absorption and utilization (Li *et al.*, 2006). The darkest yolk color score was from the DBS (10 mg/Kg and 15 mg/Kg) groups higher than that from the control. The findings are in agreement with those of Nahashon *et al.* (1994) and Mohan *et al.* (1995), who reported that a slight improvement in yolk color was observed when hens were supplemented with probiotics during the peak period of laying.

Table 2: Effects of traditional Chinese herbal medicines on egg quality index in aged laying hens

Group	N	Dosage (mg/Kg)	Yolk color	Eggshell thickness (mm)	Haugh unit	Water content of egg yolk (%)	Eggshell strength (N)
(I) Low dosage of AM	5	5	6.87±0.15	0.40±0.04	84.90±2.01	49.22±1.02	33.92±0.55
(II) Middle dosage of AM	5	10	6.81±0.17	0.41±0.02	85.29±3.17	48.12±0.61	34.09±1.01
(III) High dosage of AM	5	15	6.77±0.12	0.41±0.02	85.63±1.31	48.14±1.74	35.11±0.66
(IV) Low dosage of ASE	5	5	6.82±0.17	0.41±0.05	84.66±1.88	48.00±1.62	34.31±1.14
(V) Middle dosage of ASE	5	10	6.72±0.19	0.40±0.03	85.12±2.40	48.13±0.69	34.45±0.31
(VI) High dosage of ASE	5	15	6.96±0.08	0.41±0.01	85.41±1.43	48.61±0.80	35.23±0.41
(VII) Low dosage of DBS	5	5	6.88±0.10	0.43±0.04*	84.92±2.23	48.09±1.51	37.11±0.34*
(VIII) Middle dosage of DBS	5	10	7.25±0.11*	0.45±0.02*	87.69±1.89*	46.82±0.82*	38.21±0.81*
(IX) High dosage of DBS	5	15	7.09±0.09*	0.44±0.03*	87.48±1.31*	46.14±0.86*	37.56±0.66*
(X) control	5	0	6.75±0.14	0.40±0.04	84.69±2.19	49.57±1.18	34.02±1.05

*superscript differs notably ($P < 0.05$) compared with group X.

Table 3: Effects of traditional Chinese herbal medicines on antioxidant enzyme activities in aged laying hens

Group	N	Dosage (mg/Kg)	CAT (U/mL)	GST (ng/mL)	SOD (U/mL)	GSH (μmol/L)	MDA (nmol/mL)
(I) Low dosage of AM	30	5	17.7±2.7	3.1±0.6	456.3±21.7	368.3±28.4	7.28±0.45
(II) Middle dosage of AM	30	10	18.1±1.5	3.7±0.5	490.1±24.6	402.7±32.1	7.01±0.78
(III) High dosage of AM	30	15	16.8±2.9	3.4±0.5	507.7±33.3	374.9±11.5	6.85±0.43
(IV) Low dosage of ASE	30	5	15.3±3.8	3.7±0.3	450.2±18.9	368.2±22.4	7.59±0.58
(V) Middle dosage of ASE	30	10	17.9±1.4	3.8±0.4	461.3±24.7	359.5±21.1	7.07±1.11
(VI) High dosage of ASE	30	15	17.4±2.1	3.7±0.7	472.6±21.5	382.3±31.0	7.14±0.40
(VII) Low dosage of DBS	30	5	18.8±1.7	3.9±0.6	503.4±31.6	391.0±27.8	5.03±0.37*
(VIII) Middle dosage of DBS	30	10	19.1±1.8	5.6±4.82*	527.3±12.8	525.1±10.2*	4.23±0.59*
(IX) High dosage of DBS	30	15	16.7±2.5	3.9±4.01	481.0±23.1	355.6±12.7	6.88±0.78
(X) control	30	0	19.5±1.7	2.9±3.14	455.9±23.0	366.0±18.6	7.56±1.02

*superscript differs notably ($P < 0.05$) compared with group X.

Pharmacological actions indicated that DBT had the effects of increasing anti-oxidation activity, promoting hematopoietic functions, preventing osteoporosis, stimulating immune response (Jinous and Fereshteh, 2012; Dong, 2006; Gao, 2007). The results indicated that the GST activity were notably increased by DBS (10 mg/kg) compared with the control group. AM or ASE had no notable effects on the antioxidant enzyme activities, which may be due to the synergistic effect of Astragalus Polysaccharide and polysaccharide of Angelica sinensis. The antioxidant effect composited Ganoderma lucidum and medlar polysaccharide was stronger than the single herb polysaccharide. The antioxidant defenses include antioxidants (vitamin C, vitamin E, and uric acid) and antioxidant enzymes present in the biological system (Sies, 1991). The GST is a soluble protein that catalyzes the conjugation of GSH with many xenobiotics, and their reactive metabolites could form more water-soluble compounds (Cai and Harrison, 2000). The finding showed that expression of GST was regulated by a common antioxidant response element, which suggested that their gene products play a protective role against oxidative damage in various tissues

by neutralizing reactive oxygen species (Mates, 2000). In this experiment, the results also indicated that DBS (5 mg/kg) and (10 mg/kg) could decrease the MDA activity. The MDA endogenously reflects the lipid peroxidation (Yang *et al.*, 2008), which is the consequence of diminished antioxidant protection when levels of reactive oxygen species increase (Yang *et al.*, 2008). All together, the results demonstrated that supplementation of DBS (10 mg/Kg) has obviously impact on the antioxidant enzymes activities in laying hens. The results indicated that DBS (10 mg/kg) could improve the antioxidant enzyme activities, egg production, egg quality. So it was suitable for the clinical manifestations. The results were the implication to chickens in several farms for two years. It increased egg production and laying rate and reduced medical expenses due to diseases, which will bring huge economic and social benefits. One of the limitations of this study is that there was no research on the stability of antioxidant enzyme activities of DBS, but with the development of scientific, technological and traditional Chinese medicine theory, the stability and mechanism of DBS on improving antioxidant enzyme activities will be deepened and systematized gradually.

REFERENCES

- Beers RF and Sizer IW (1952). A spectrophotometric method for measuring the breakdown of hydrogen peroxide by catalase. *J. Biol. Chem.* **195**:133–140.
- Cai H and Harrison DG (2000). Endothelial dysfunction in cardiovascular diseases: The role of oxidant stress. *Circ. Res.* **87**:840–844.
- Cho WC and Leung KN (2007). In vitro and in vivo antitumor effect of Astragalus membranaceus. *Cancer. Cel.* **252**:430.
- Dang SS, Jia XL, Cheng YA, Chen YR, Liu EQ and Li ZF (2004). Inhibitory effect of Huangqi Zhechong decoction on liver fibrosis in rat. *World. J. Gastroenterol.* **10**:2295–2298.
- Dong TTX, Zhao KJ, Gao QT, Ji ZN, Zhu TT, Li J, Duan R, Cheung AW and Tsim KW (2006). Chemical and biological assessment of a Chinese herbal decoction containing Radix Astragali and Radix Angelicae Sinensis: determination of drug ratio in having optimized properties. *J. Agr. Food. Chem.* **54**:2767–2774.
- Gao QT, Cheung JKH, Li J, Jiang ZY, Chu GK, Duan R, Cheung AW, Zhao KJ, Choi RC, Dong TT and Tsim KW (2007). A Chinese herbal decoction, Danggui Buxue Tang, activates extracellular signal-regulated kinase in cultured T-lymphocytes. *Febs. Lett.* **581**:5087–5093.
- Gao Q, Li J, Cheung JKH, Duan J, Ding A, Cheung AW, Zhao K, Li WZ, Dong TT and Tsim KW (2007). Verification of the formulation and efficacy of Danggui Buxue Tang (a decoction of Radix Astragali and Radix Angelicae Sinensis):

- an exemplifying systematic approach to revealing the complexity of Chinese herbal medicine formulae. *Chin. Med.* **2**:5077-5088.
- Gomez E, Irvine DS and Aitken RJ (1998). Evaluation of a spectrophotometric assay for the measurement of malondialdehyde and 4-hydroxyalkenals in human spermatozoa: Relationships with semen quality and sperm function. *Int. J. Androl.* **21**:81-94.
- Jinous A and Fereshteh R (2012). Chemistry, pharmacology and medicinal properties of *Peganum harmala* L. *Afr. J. Pharm. Pharmacol.* **6**:1573-1580.
- Liang Y, Lu XH and Chen RQ (1999). Research progress of blood-deficient syndrome in traditional Chinese Medicine. *Chin. J. Inf. TCM.* **6**:16-18.
- Li Ch, Dong Yl, Hou Hf, Li Q, Zhang Rh, Qin Rl, Li Zh, BaoYz and Shi Wy (2013). Effects of traditional Chinese herbal medicines on blood cell count and immunity in chickens. *Afr. J. Pharm. Pharmacol.* **7**:2081-2086.
- Li J, Bao X, Zhu X, Liu J and Wang HP (2008). Immunoregulatory activity of polysaccharopeptide and Astragalus polysaccharides on EAC tumor-bearing mice. *Zhongguo Zhongyao Zazhi* **33**:924-927.
- Li L, Xu CL, Ji C, Ma Q, Hao K, Jin ZY and Li K (2006). Effects of a dried *Bacillus subtilis* culture on egg quality. *Poult. Sci.* **85**:364-368.
- Mates JM (2000). Effects of antioxidant enzymes in the molecular control of reactive oxygen species toxicology. *Toxicology* **153**:83-104.
- Mohan B, Kadirvel R, Bhaskaran M and Natarajan A (1995). Effect of probiotic supplementation on serum/yolk cholesterol and on egg shell thickness in layers. *Br. Poult. Sci.* **36**:799-803.
- Nahashon SN, Nakaue HS and Mirosh LW (1994). Production variables and nutrient retention in single comb White Leghorn laying pullets fed diets supplemented with direct-fed microbials. *Poult. Sci.* **73**:1699-1711.
- Palmer SS and Bahr JM (1992). Follicle stimulating hormone increase serum oestradiol-17 beta concentration, number of growing follicles and yolk deposition in aging hens (*Gallus domesticus*) with decreased egg production. *Br. Poult. Sci.* **33**:403-414.
- Resta-Lenert S and Barrett KE (2003). Live probiotics protect intestinal epithelial cells from the effects of infection with enteroinvasive *Escherichia coli* (EIEC). *Gut* **52**:988-997.
- Shang P, Qian AR, Yang TH, Jia M, Mei QB, Cho CH, Zhao WM and Chen ZN (2003). Experimental study of anti-tumor effects of polysaccharides from *Angelica sinensis*. *World. J. Gastroenterol.* **9**:1963-1967.
- Shao P, Zhao LH, Zhi-Chen and Pan JP (2006). Regulation on maturation and function of dendritic cells by Astragalus mongholicus polysaccharides. *Int. J. Immun.* **6**:1161-1166.
- Sies H (1991). Oxidative stress: From basic research to clinical application. *Am. J. Med.* **91**:31S-38S.
- Sima N, Mahdiah A and Narges K (2012). Evaluation of analgesic and anti-inflammatory effects of fresh onion juice in experimental animals. *Afr. J. Pharm. Pharmacol.* **6**:1679-1684.
- Tsai NM, Lin SZ, Lee CC, Chen SP, Su HC, Chang WL and Harn HJ (2005). The antitumor effects of *Angelica sinensis* on malignant brain tumors in vitro and in vivo. *Clin. Cancer. Res.* **11**:3475-3484.
- Winterbourn CC, Hawkins RE, Brain M and Carell R (1975). The estimation of red cell superoxide dismutase activity. *Journal of Laboratory and Clinical Medicine*, **85**: 337-341.
- Yang Q, Populo SM, Zhang J, Yang G, Kodama H (2002). Effect of *Angelica sinensis* on the proliferation of human bone cells. *Clin. Chim. Acta.* **324**:89-97.
- Yang RL, Li W, Shi YH and Le GW (2008). Lipoic acid prevents high-fat diet-induced dyslipidemia and oxidative stress: A microarray analysis. *Nutrition* **24**:582-588.
- Yuan C, Pan X, Gong Y, Xia A, Wu G, Tang J and Han X (2008). Effects of Astragalus polysaccharides (APS) on the expression of immune response genes in head kidney, gill and spleen of the common carp, *Int. Immunopharmacol.* **8**:51-58.
- Zakowski JJ and Tappel AL (1978). A semiautomated system for measurement of glutathione in the assay of glutathione peroxidase. *Analytical Biochemistry*, **89**: 430-436.